

Pratham Chauhan

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Professional Summary

Results driven Computer Science graduate from Rutgers University with proven experience building full-stack applications, ML pipelines, and cloud-backed systems. Shipped production-ready projects using Java, Python, JavaScript, React, Node.js, FastAPI, and AWS. Managed IT infrastructure for 9,000+ daily users with zero critical failures. Adept at translating complex technical problems into clean, scalable solutions across software engineering, data engineering, and systems roles.

Technical Skills

Languages: Java, Python, C, C++, C#, JavaScript, SQL, Swift

Web & Backend: Node.js, Express.js, REST API Development, React, HTML, CSS, API Integration

Databases: MySQL, MongoDB, JDBC, Core Data, SQL Query Optimization, Data Modeling

AI / ML: Machine Learning, Generative AI, OpenAI APIs, scikit-learn, pandas, NumPy, Jupyter Notebook

Cloud & DevOps: AWS (EC2, S3), Git, GitHub, Bash, Linux/Unix, CI/CD, Docker

Mobile: Swift, SwiftUI, UIKit, Core Data, Xcode

Methodologies: Agile, OOP, SDLC, Unit Testing, Debugging, Technical Documentation, Object-Oriented Design

Education

Bachelor of Science, Computer Science

May 2026

Rutgers University – New Brunswick, NJ

GPA: 3.9 | Dean's List

- Coursework: Data Structures & Algorithms, Data Science, Internet Technology, Systems Programming, Computer Security, Principles of Programming Languages.

Associate of Science, Computer Science

Jun 2024

Middlesex College – Edison, NJ

GPA: 3.8 | Dean's List (All Semesters)

- Coursework: Java, OOP, Data Structures, Discrete Mathematics, Computer Architecture, Calculus I & II | Phi Theta Kappa Honor Society.

Experience

Computer Science Teaching Assistant

Sep 2025 – May 2026

Rutgers University – Design & Analysis of Algorithms

New Brunswick, NJ

- Developed standardized grading rubrics with faculty and evaluated 50+ programming assignments per cycle across 200+ students, ensuring consistent and objective assessment aligned with course learning outcomes.
- Delivered targeted debugging sessions on dynamic programming, graph traversal, and divide-and-conquer algorithms, reducing recurring logic errors in submitted code.
- Authored LaTeX solution guides covering max-flow, NP-hardness reductions, ILP formulations, and Kruskal's algorithm; crafted Socratic discussion-board responses to guide students without revealing answers.

Building Operations Manager

Apr 2025 – May 2026

Rutgers University

New Brunswick, NJ

- Managed IT infrastructure for a 61,000 sq. ft. facility serving 9,000+ daily visitors across 30+ Windows/Linux workstations; diagnosed and resolved hardware, software, and networking failures.
- Configured AV and network infrastructure for 100+ annual events, achieving zero critical technical failures through pre-event audits, real-time monitoring, and documented incident-response procedures.

Projects

Spending Insights Dashboard | *React, Vite, FastAPI, Python, pandas, NumPy, scikit-learn, Recharts*

Apr 2026

- Built a full-stack financial analytics dashboard ingesting raw bank transaction CSVs and transforming them into interactive spending breakdowns, monthly trend charts, anomaly detection alerts, and ML-backed next-month spend forecasts.
- Implemented a FastAPI backend with pandas/NumPy data pipelines and a React/Recharts frontend with demo-data loading, dark presentation mode, and printable PDF report export.

Fashion Trend Analysis & Recommendation System | *Python, pandas, NumPy, scikit-learn, Machine Learning*

Dec 2025

- Built a data-driven ML pipeline to analyze and predict emerging fashion trends; engineered feature extraction and preprocessing workflows using pandas and NumPy.
- Trained and iteratively optimized scikit-learn classification models via cross-validation and hyperparameter tuning, improving prediction accuracy on held-out test sets.

SmartPantry – AI Recipe Generator (HackRU 2025) | *Node.js, Express.js, MongoDB, REST API, OpenAI API*

Feb 2025

- Developed and deployed a full-stack web app integrating barcode scanning and object-recognition APIs to deliver personalized recipe recommendations; demoed live at HackRU 2025.
- Implemented RESTful API endpoints, optimized MongoDB schemas for scalable data modeling, session-based authentication, and a responsive frontend UI.

Certifications

Developing Generative AI Solutions – AWS | **Fundamentals of Machine Learning & AI – AWS**

Programming in C# | **Programming in C++** | **Python Data Structures & Algorithms**